Tanuj Dargan

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Education & Awards

University of Victoria – International Scholarship Victoria, BC
B.S. Computer Science – Accelerated 3-Year Completion *Apr. 2027*
Relevant Coursework: Quantum Computing, Computer Architecture, Numerical Analysis, Communications & Networks

Work Experience

Amazon Robotics May 2026 – Aug 2026
Software Development Engineer Intern *Toronto, ON*

- Shipped an LLM-powered planning assistant to production in Optimus, the labor optimization platform for **180+ Amazon fulfillment sites across NA and EU**, replacing manual calculations with natural-language plan queries, anomaly detection, and labor recommendations; projected to save **$136.2M annually** via reduced operator-hours wastage and optimal lever overrides.
- Built a **RAG pipeline** (vector search + prompt-engineered chains) over historical labor plans and site configurations to contextualize responses to site-specific parameters (ARS, TNS), **cutting planner time-to-insight by 60%+**.
- Designed a **multi-turn interface** on Optimus' optimization engine for **what-if labor scenario simulation** and natural-language constraint adjustment, returning **real-time re-optimized plans** without full solver re-runs.

RBC Borealis – AI Mentorship Program Mar. 2026 – May. 2026
AI/ML Research Mentee *Remote*

- Engineered an **ML pipeline flagging at-risk youth** navigating chronic illness by training PyTorch models on **1,500+ longitudinal engagement records (3+ years, 20+ behavioural features)**, isolating **4+ user cohorts via UMAP clustering** to inform model design and enable earlier clinical intervention.
- Designed an analytics dashboard serving **4+ clinical and program teams**, visualizing retention curves, engagement trends, and **disengagement risk flags** with Python, Matplotlib, and GA4.

Pear Care Jul. 2025 – Apr. 2026
Lead AI Developer *Remote*

- Accelerated diagnostic workflows by building **autonomous medical-AI agents** with LangGraph, **MoE routing**, and task-introspective reasoning, served via FastAPI on GCP across **1,000+ clinical specialties**.
- Cut **model memory footprint by 40%** while preserving **95%+ F1** on internal clinical triage benchmarks by self-hosting with PyTorch, Ollama, and vLLM, then transitioning to **quantized LoRA/PEFT** fine-tuned deployments.
- Architected for **horizontal scale** across a multi-tenant **GCP/AWS infrastructure** with Redis caching, DynamoDB persistence, **end-to-end encryption**, and mobile-first access, validated through load and stress testing.

Major League Hacking (MLH) SWE Fellowship - Apache Airflow May. 2025 – Aug. 2025
Software Engineering Fellow *Remote*

- Streamlined developer debugging for Apache Airflow's **50K+ star codebase** by architecting **Docker-based IDE integration** (VSCode, PyCharm) with step-through debugging and dynamic port exposure, cutting **setup-to-debug cycles from hours to minutes**; shipped in a team of 3 alongside Royal Bank of Canada engineers.

DEIA (Database & Data Mining) May. 2025 – Feb. 2026
Undergraduate Researcher - Prof. Sean Chester *Victoria, BC*

- Built a custom retrieval system on a modified cuVS fork (CAGRA, IVF-PQ) sustaining **1–4M QPS at billion-vector scale** with **90–95% recall** and **sub-250 ms latency** across 8+ NVIDIA GPUs and 2TB of indexed data.
- Achieved **4.3× throughput scaling** by building automated stress-testing pipelines on **Kubernetes GPU clusters**, profiling memory bandwidth and **NVLink interconnect saturation** to expose bottlenecks.

UVic Centre for Aerospace Research (CFAR) Apr. 2025 – Apr. 2026
Software Developer & Undergraduate Researcher *Victoria, BC*

- Led **OBC firmware (C), TTC board design**, and **cross-functional hardware/software integration** for **MarmotSat, now in orbit** after launching aboard a **SpaceX Falcon 9**, coordinating across electrical, mechanical, and software teams.
- Cut testing from **180 min to 15 min** by building a **CI/HIL pipeline** with GitHub Actions and SWD flashing (80+ checksum rules/commit), and implementing C/Java YAMCS integration that eliminated **20+ hours of manual work per release**.

Projects

Apart Digital Minds Research Sprint August 2026

- Demonstrated that loss-masking regimes during SFT determine model distress vocalization: the assistant-only recipe maximizes expressed shutdown distress, while any loss on user turns restores a neutral register without general capability degradation.
- Found an expression-representation dissociation via valence probing: models fine-tuned to sound calm register shutdown negatively before generation, meaning a calmer-sounding model is not a better-off one and output-based welfare signals are unreliable.

Optimate: Hack the North 2025 Winner | *Next.js, Cohere, AWS DynamoDB* Sep. 2025

- Won **2 prizes out of 300+ teams** (**YC Unicorn Prize** & **Federato Best RiskOps**) by building an AI underwriting dashboard that automated risk assessment with Next.js, Cohere LLMs with RL, and DynamoDB; Federato CEO validated **40% faster review cycles** and **15% fewer underwriting errors** versus manual workflows.

AIMO Progress Prize 2 Competitor: Silver Medal | *Finishing score: 27* Nov. 2024 – Mar. 2025

- Achieved **54% accuracy**, finishing **top 10% of 2,200 teams** by engineering **multi-stage LLM pipelines** with fine tuned open weight models, NEMO skills reasoning strategies and ablation studies for in-context learning.